

```
public class Item {
    private String userId;
    private String itemName;
    private int itemUnitPrice;
    private int quantity;

    public Item(String uid, String itemName, int itemUnitPrice) {
        this.userId = uid;
        this.itemName = itemName;
        this.itemUnitPrice = itemUnitPrice;
        this.quantity = 1;
    }

    public Item(String uid, String itemName, int itemUnitPrice, int quantity) {
        this.userId = uid;
        this.itemName = itemName;
        this.itemUnitPrice = itemUnitPrice;
        this.quantity = quantity;
    }

    public void upQuantity() {

        this.quantity += 1;

    }

    public void downQuantity() {
        this.quantity--;
    }

    public String getUserId() {
        return this.userId;
    }

    public String getItemName() {
        return this.itemName;
    }

    public int getItemUnitPrice() {
        return itemUnitPrice;
    }
}
```

```
    public int getQuantity() {  
        return quantity;  
    }  
}
```

```
public class Node {
```

```
    Node next;
```

```
    Item data;
```

```
    Node(Item data) {  
        this.data = data;  
        this.next = null;
```

```
    }
```

```
}
```

```
public class LinkedList {
```

```
    Node head;
```

```
    Node tail;
```

```
    LinkedList() {  
        this.head = null;  
        this.tail = null;
```

```
    }
```

```
    public void insertItem(String uid, String itemName, int itemUnitPrice) {
```

```
        Item item = new Item(uid, itemName, itemUnitPrice);
```

```
        Node newNode = new Node(item);
```

```
        if (head == null) {  
            head = newNode;  
            tail = newNode;
```

```
        } else {
```

```
            if (searchItem(uid, itemName) != null) {  
                this.searchItem(uid, itemName).data.upQuantity();
```

```

        } else {
            tail.next = newNode;
            tail = newNode;

        }

    }
}

public void insertItem(String uid, String itemName, int itemUnitPrice, int quantity) {
    Item item = new Item(uid, itemName, itemUnitPrice, quantity);
    Node newNode = new Node(item);
    if (head == null) {
        head = newNode;
        tail = newNode;

    } else {

        tail.next = newNode;
        tail = newNode;

    }

}

public boolean nodeExists(String uid, String itemname) {
    Node current = head;
    while (current != null) {
        if (current.data.getItemName() == itemname) {
            return true;
        }
        current = current.next;

    }
    return false;
}

public Node searchItem(String uid, String itemname) {
    Node current = head;
    if (nodeExists(uid, itemname)) {
        while (current != null) {

```

```
        if (current.data.getItemName().equals(itemname) &&  
current.data.getUserId().equals(uid)) {
```

```
            return current;  
        }  
        current = current.next;  
    }  
}
```

```
    }  
    return null;
```

```
}
```

```
// to decrease
```

```
void updateQuantity(String uid, String item) {
```

```
    searchItem(uid, item).data.downQuantity();  
}
```

```
/*  
 * we can increase the quantity by adding a new item  
 * or  
 *  
 * void increaseQuantity(String uid, String item){  
 *  
 * searchItem(uid, item).data.upQuantity();  
 * }  
 *  
 *  
 */
```

```
void removeItem(String uid, String item) {
```

```
    Node temp = head, prev = null;  
    if (temp == searchItem(uid, item)) {  
        head = temp.next;  
        return;  
    }
```

```
    while (temp != searchItem(uid, item)) {  
        prev = temp;  
        temp = temp.next;  
    }
```

```
    if (temp == null) {  
        return;
```

```

    }

    prev.next = temp.next;

}

public LinkedList filterByUser(String uid) {

    Node temp = head;
    LinkedList user = new LinkedList();

    while (temp != null) {
        if (temp.data.getUserId().equals(uid)) {
            user.insertItem(temp.data.getUserId(), temp.data.getItemName(),
temp.data.getItemUnitPrice(),
temp.data.getQuantity());
        }

        temp = temp.next;
    }

    return user;
}

public void printUser(String uid) {

    LinkedList user = filterByUser(uid);

    System.out.println("User ID : " + uid);

    Node temp = user.head;

    int i = 1;
    while (temp != null) {
        System.out.println(i + ".");
        System.out.println("Item Name : " + temp.data.getItemName());
        System.out.println("Item Price : " + temp.data.getItemUnitPrice());
        System.out.println("Item Quantity : " + temp.data.getQuantity());

        temp = temp.next;
        i++;
    }
}

```

```

    }

}

public void calTheBill(String uid) {
    LinkedList user = filterByUser(uid);
    Node temp = user.head;
    int total = 0;
    while (temp != null) {
        total += temp.data.getItemUnitPrice() * temp.data.getQuantity();
        temp = temp.next;
    }
    System.out.println("Total Price : " + total);
}

}

}

```

```

public class Driver {
    public static void main(String[] args) {
        LinkedList list1 = new LinkedList();

        list1.insertItem("user1", "Book", 100);

        list1.insertItem("user1", "pen", 200);
        list1.insertItem("user1", "pen", 200);
        list1.insertItem("user1", "bottle", 500);
        list1.insertItem("user1", "pen", 200);
        list1.insertItem("user2", "pencil", 400);
        list1.insertItem("user2", "Book", 100);
        list1.insertItem("user3", "bottle", 500);
        list1.insertItem("user3", "Book", 100);
        list1.insertItem("user1", "Book", 100);
        list1.insertItem("user1", "Book", 100);

        list1.printUser("user1");
        System.out.println("#####");
        // remove Item completely

        list1.removeItem("user1", "pen");
        list1.printUser("user1");
    }
}

```

```
System.out.println("#####");
// decrease item count
list1.updateQuantity("user1", "Book");

list1.printUser("user1");
System.out.println("#####");

list1.calTheBill("user1");

}
}
```